

Trust Formation in University–Industry Collaborations in the U.S. Biotechnology Industry: IP Policies, Shared Governance, and Champions^{*,†}

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Haggling over rights to potential inventions can be a major roadblock to successful university–industry (UI) collaborations. Yet such collaborations are critical for innovation in science-based industries. This study examines the roles of universities' intellectual property (IP) policies and of shared governance for trust formation between academe and industry. The study also examines how UI champions moderate this process and how trust between university and industry partners affects UI collaboration outcomes. The analysis of survey data of 105 recent UI collaborations in the U.S. biotechnology industry indicates that the flexibility and transparency of university IP policies and shared governance by UI partners are both positively related to trust formation. The activities of UI champions amplify the positive effects of shared governance and at the same time reduce the importance of university IP policies for trust formation between UI partners. The amount of trust between partners is positively related to knowledge transfer and innovation performance. The findings suggest that despite widely reported industry concerns over the control of IP, UI research partners can develop a trustful environment and thereby plant the seeds for a successful collaboration. In order to enhance trust, companies should not only consider university IP policies, but also need to actively engage in shared governance with university partners. UI collaboration champions can help shift the attention of company managers from formal rules set by university IP policies toward shared project planning, coordination, and implementation with university partners.

[Pfizer] is betting that it can bring more treatments to market by working with academic researchers in medical hubs like Boston . . . Overall, it is committing up to \$100 million to such collaborations over the next five years.

Boston Globe, July 30, 2012

I think academic industry collaboration is the engine for future growth of technology and ultimately the wealth of nations. You have to remain cutting edge and there has to be investment.

Interview with industry partner

Introduction

University–industry (UI) collaborations for the research and development (R&D) of new products and technologies are an increasingly important way in which firms can obtain, combine, and leverage their R&D resources (Leydesdorff and Meyer, 2006; Sherwood and Covin, 2008). Because of a reduction of industrial basic research laboratories both in number and size, industry has come to rely increasingly on academic laboratories for research output. At the same time, productive and well-established research relations with industry are important for those universities seeking to possess a vibrant and dynamic research enterprise.

The convergence of these circumstances results in unprecedented opportunities for successful partnerships between academe and industry particularly in science and technology-based industries such as biotechnology. In these industries, basic research is critical for innovativeness and the long-term success of companies (Dosi, 1988; Welsh, Glenna, Lacy, and Biscotti, 2008). However, universities and companies face major challenges when attempting to work together. The challenges

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partially originate in different organizational cultures, schedules, agendas, or management styles (Bower, 1992; George, Zahra, and Wood, 2002). Often these differences are exacerbated through difficulties in negotiating collaborative research agreements (Burnside and Witkin, 2008), resulting in frustration and mistrust for all parties involved and leading to delays or cancellations of joint research projects. Thus, realizing the proposed benefits of such UI collaborations often turns out to be very challenging.

Many of the impediments to UI collaborations are related to the treatment of intellectual property (IP). The often difficult and lengthy negotiations about IP are a significant barrier and a continual challenge in the development of new collaborations and, according to some, a barrier to American competitiveness (Burnside and Witkin, 2008). Universities are intent on protecting the IP they have built up over decades, whereas companies need to be allowed to use the IP generated in UI collaborations to build their future business on it. In many cases, the cost of putting together a research project and negotiating the terms may be greater than the value of the proposed research. As a result, some U.S. companies have given up talking with U.S. universities in frustration and instead have worked out arrangements with foreign universities with more favorable arrangements (Kramer, 2008). Moreover, companies also typically insist on delaying

publication of research results so they can comb the results for potentially patentable material, a restriction that runs counter to the free and open dissemination of the results pursued by university researchers (Kramer, 2008). These realities suggest that balanced IP policies are crucial for effective UI collaborations.

A second barrier to effective UI collaborations relates to the governance of the project as the collaboration unfolds. Perkmann and Salter (2012) argue that UI collaborations, particularly long-term ones, require careful attention to “the design and governance of the collaboration” (p. 80). The collaboration needs to balance the firm’s desire for practical results that will enable their business with the university researchers’ orientation toward academic knowledge development (Perkmann and Salter, 2012). Thus, shared governance, in which both parties jointly determine the focus of the collaboration, carry mutual responsibility for its progress, and together address issues that arise over the course of the relationship, can help create an effective UI collaboration (Burnside and Witkin, 2008). As well, both partners need to be flexible in their negotiations and expectations (Gopalakrishnan and Santoro, 2004; Jelinek and Markham, 2007).

In this study, a UI research collaboration is defined as an R&D-based relationship involving a private firm and a university that are mutually committed to reaching a common R&D goal by pooling their resources or coordinating specific research activities. To enhance our understanding of productive UI collaborations, this research examines how university IP policies and shared governance facilitate trust formation and what role UI champions play in this process. IP policies and shared governance are potentially significant facilitators of a trusting relationship and ultimately successful outcomes in UI collaborations (Burnside and Witkin, 2008; Gopalakrishnan and Santoro, 2004; Kramer, 2008). Trust can be defined as a psychological state comprising the intention to accept vulnerability based upon one party’s positive expectations of the intentions or behavior of another party in situations that are interdependent or risky (Morgan and Hunt, 1994; Rousseau, Sitkin, Burt, and Camerer, 1998). When partners trust each other, they can focus on the R&D challenges at hand instead of devoting precious time and efforts to haggling over rights to potential inventions or on the process for dealing with unexpected issues.

Contract negotiation, flexible arrangements, and ownership and protection of IP are extremely important in the biotechnology industry. The biotechnology industry is still relatively nascent (Nelson, 1991) and more reliant on

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research from universities than even the pharmaceutical business. In fact, most of the basic research in the biotechnology industry is done in university laboratories (McMillan, Narin, and Deeds, 2000). Thus, examining the issues of IP policies and shared governance within the biotechnology industry should enhance our understanding of how to create effective UI collaborations.

The contributions of this research are threefold. First, most empirical research suggests that large firms have a greater tendency to pursue UI collaborations (Schartinger, Rammer, Fischer, and Fröhlich, 2002; Veugelers and Cassiman, 2005) because they have larger R&D resources that allow them to access external sources of expertise (Cohen, 1996). Howells, Ramlogan, and Cheng (2012) found that larger firms rate universities higher than smaller firms as sources of information and knowledge. However, Eom and Lee (2010) argue that small firms may be more likely to engage in UI collaborations because they lack the R&D resources (financial, people, equipment) to do the research themselves. That is, small firms, particularly those in high tech industries, may need to collaborate with universities because of the limited nature of their R&D capabilities and the type of industry within which they compete (Veugelers and Cassiman, 2005). Cosh and Hughes (2010) found that small and medium firms in the United States used universities as sources of knowledge more than large firms. By focusing on small and medium biotech firms, this research will provide greater insight into small and medium-sized enterprises' (SMEs) involvement in UI collaborations in the biotechnology industry.

Second, there is a dearth of research that examines the role of trust in UI collaborations. Although a few studies have investigated trust (cf. Bruneel, D'Este, and Salter, 2010; Gopalakrishnan and Santoro, 2004; Santoro and Saporito, 2003), only Santoro and Saporito (2003) put trust at the center of their conceptual model. Taking a similar approach, the current study focuses on the firm's trust in its university partner. Additionally, this research explores the relevance of IP policies, shared governance, and champion behavior for facilitating trust. As well, the research addresses the relationship between trust and substantive UI collaboration outcomes. Emphasis on the latter fills a gap identified in the literature. Specifically, Jelinek and Markham (2007) indicate that not enough is known about the effectiveness of UI collaborations. Similarly, Veugelers and Cassiman (2005) state that there is a need to address "the efficiency of cooperation and its impact on innovation performance and growth" (p. 374).

Finally, by exploring the moderating effect of a UI champion on the impact of IP policies and shared gover-

nance on trust, this study provides greater understanding of how a champion may facilitate the development of trust. While the role of champions as important facilitators of innovation outcomes has been advanced in the literature on product development teams (e.g., Chakrabarti, 1974; Howell and Shea, 2006), or on university spin-offs (Walter, Parboteeah, Riesenhuber, and Hoegl, 2011), it has received little attention as a key facilitator of successful UI collaborations (Santoro and Chakrabarti, 2002). This is surprising given the importance of the champion as the key liaison between the two very different mind-sets and operating philosophies of the university and industry partners.

Conceptual Framework

Collaborations between firms and universities are likely when the former need access to specialized knowledge that they do not possess. Academic institutions can provide expert technological knowledge, particularly in the early stages of the innovation process, that can lead to new or breakthrough products. Thus, certain firms, such as those in the biotechnology industry, have a strong interest in the scientific and technological knowledge offered by universities (Veugelers and Cassiman, 2005).

Governance mechanisms in UI collaborations are necessary to clarify initial collaboration conditions, facilitate the agreement between parties to perform particular actions, allow adjustments to technological and market changes, and in general to reduce behavioral uncertainty (Gopalakrishnan and Santoro, 2004). Behavioral uncertainty can also be reduced by closely coordinating work and adjusting plans when necessary (Macneil, 1978). Thus, this study examines shared governance and IP policies as mechanisms in UI collaborations that can enhance trust between the partners.

The level of trust is an important factor in R&D collaborations, particularly between partners with little prior knowledge about each other (Christiansen and Vendelø, 2003). Effective governance also requires strong leadership and flexible and skilled champions committed to the UI collaboration to foster a comfort level between the parties involved and to prevent potential breakdowns or terminations, should unanticipated situations arise (Barnes, Pashby, and Gibbons, 2002; Prabhu, 1999; Santoro and Betts, 2002; Santoro and Chakrabarti, 2002). Thus, the relationships between IP policies and shared governance with trust formation are likely to be contingent upon skilled and committed UI champions. Creating a trusting atmosphere, in turn, will facilitate positive

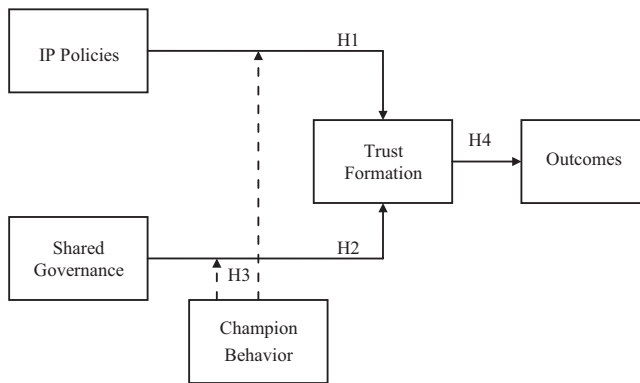


Figure 1. Conceptual Framework

outcomes amplifying the joint R&D of new products and technologies (see Figure 1).

IP Policies and Trust Formation

IP policies are defined as rules and measures set up by universities to protect IP created by university researchers, maximize revenues, and facilitate knowledge transfer (Santoro and Bierly, 2006). Many firms view the universities' demand for exclusive ownership of IP rights as overly excessive and one-sided and thus as an impediment for working with universities (Santoro and Betts, 2002). Others criticize universities for employing the same IP policy approach for very different industries (Kramer, 2008) and for frustrating their industry partners by bureaucratic or inflexible behavior, such as rigidly insisting on exclusive ownership of patents and other IP or on their right to immediately publish research outcomes without any restriction or delay (Santoro and Bierly, 2006). Such demands can inhibit the development of productive working relationships between universities and firms, as the firms' managers gain the perception that the university partner is unresponsive to their needs. Moreover, when the universities' IP policies are not clear and transparent, there is a high risk that conflicts between firms and universities will arise later on regarding IP ownership or the dissemination of results.

Conversely, when universities acknowledge there is no "one-size-fits-all-approach" and are flexible to tailor IP agreements, firms show a higher propensity to commit more time and resources to the partnership (Gopalakrishnan and Santoro, 2004), as they feel that their concerns are being addressed by the university partner. The more clear, transparent, and fair the universities' IP policies are perceived by company partners, the less room there is for ambiguity and future conflicts over IP issues. Since IP policies are typically a major source of

conflict within UI collaborations (Jelinek and Markham, 2007), such openness and transparency can induce both sides to share important and sensitive information (Santoro and Bierly, 2006). Disclosing information can help each party better understand the other's capabilities, expectations, and ways of doing things (Santoro and Saporito, 2003). As well, clear and equitable IP policies can reduce uncertainty about the partner's future behavior. Thus, flexible and transparent IP policies by universities can facilitate trust with industry partners.

H1: Flexible and transparent university IP policies are positively related to trust formation in UI collaborations.

Shared Governance and Trust Formation

Representatives from different organizational cultures are used to their own ways of organizing their work and problem solving. These differences can easily lead to unproductive friction and conflict and negatively affect trust formation. At the same time, the context within which the collaboration takes place may change. In such cases, it is essential that both parties work together to adjust their mutual agreements to the changed environment by means of relational governance (Macneil, 1978). Engaging in such shared adaptive efforts helps to overcome interaction and coordination barriers and increases flexibility and cooperation in UI collaborations.

Shared governance is defined as the degree to which the collaboration partners share the responsibility for maintaining the collaboration itself and for jointly responding to problems that arise as the project goes on (Heide and Miner, 1992). It involves task-related and ongoing mutual efforts and decisions that the counterparts undertake to diagnose or overcome obstacles blocking project effectiveness (Narus and Anderson, 1995), including adaptations to initial agreements that did not account for every eventuality. Shared governance can reduce information asymmetry (Saxton, 1997), promote a common understanding of goals, and help prevent misunderstandings (Asmus and Griffin, 1993). Coordinating goals and skills between partners can also be a powerful socialization mechanism in forging effective collaboration and teamwork (Lawson, Peterson, Cousins, and Handfield, 2009). In other words, shared governance can be a bonding mechanism that helps partners adapt to new circumstances, facilitates the sharing and integration of information, and enables the breeding of positive expectations about how a productive working relationship can be developed within a changing environment. The result of shared governance is trust formation as there is less uncertainty about a partner's intentions and behaviors

because both parties collectively work through project challenges, negotiate mutual adaptations, and make project-related decisions.

H2: Shared governance is positively related to trust formation in UI collaborations.

The Moderating Effect of Champion Behavior

University IP policies and shared governance can help clarify the initial collaboration framework and reduce uncertainty between a university and the company partner by providing clarity and a common base of mutual understanding. Ultimately, however, successful collaborations within the context of governance mechanisms depend on individual leadership efforts, as collaborative research is often met by ignorance, skepticism, and resistance, particularly between academe and industry (Bruneel et al., 2010; Santoro and Chakrabarti, 2002). In this regard, innovation champions play a crucial role for the success of UI collaborations. The importance of a champion lies in promoting a collaborative idea to the firm and getting management sufficiently interested in the project so that the firm will commit to it and provide the necessary financial support, and in acting as firefighters in cases of emergencies.

Here, a UI champion is defined as “an individual who is intensely interested and involved with the overall objectives and goals of the project and who plays a dominant role in many of the research-engineering interaction events through some of the stages, overcoming technical and organizational obstacles, and pulling the effort through its final achievement by the sheer force of his or her will and energy” (Chakrabarti, 1974, p. 58). The industry champion has been found to be more critical to the collaboration than the university champion (Santoro and Chakrabarti, 2002).

Champions within the firm shepherd and facilitate negotiating the initial agreement with the university, secure initial funding for a joint project, solicit top management commitment, and get the project off the ground. They also deal with challenges that arise during the collaboration, persuade opponents to stop their resistance to the project, and secure financial and top-level managerial support when necessary, thus enabling the collaborators to work effectively on the project (Howell and Shea, 2006).

Research shows that champions indirectly affect collaboration outcomes by facilitating social relationships and bringing people together, who then will also have a basis for communication on other issues (Hewes and Lyons, 2008; Markham and Griffin, 1998). They can redi-

rect the attention of key actors from the formal collaboration framework toward the actual interaction with collaboration partners. UI champions can help separate the legal framework of initial IP considerations from the more important task of facilitating a mutually productive working relationship (Burnside and Witkin, 2008). It is often impossible to determine outcomes or place fair values on potential outcomes early on in the collaboration (Jelinek and Markham, 2007). As the collaboration progresses, initially agreed upon IP agreements are likely to shift as these may not account for new developments or unexpected results of an ongoing project. Such developments will benefit from the intervention of champions, which can alleviate concerns over deliverables and facilitate and encourage collaboration activities (Santoro and Chakrabarti, 2002). At the same time, UI champions can persuade collaboration partners to adapt and redefine their working goals and plans over the course of the project as needs and environments change. The quality of these adaptive exchanges will affect perceptions as to the value of the progressing UI collaboration (Inkpen and Tsang, 2005), which in turn, will determine whether the partners can develop and maintain a trusted relationship or not.

Taken together, effective UI champions are likely to shift the focus of attention of company managers involved in a UI collaboration away from the formal rules set by university IP policies and redirect it toward shared project planning, coordination, and implementation with university partners. As a consequence, IP policies may become less relevant, and shared governance may become more relevant to trust formation in the presence of effective champion behavior.

H3a: Higher levels of champion behavior weaken the relationship between IP policies and trust.

H3b: Higher levels of champion behavior strengthen the relationship between shared governance and trust.

Trust Formation and Collaboration Outcomes

Trust has been found to serve as the glue that cements UI collaborations and creates a climate that facilitates joint activities and positive outcomes (Gopalakrishnan and Santoro, 2004). When sharing project and problem-solving responsibilities, collaboration partners will be better able to understand and absorb knowledge from each other (Saxton, 1997). Research into collaborations faced with high uncertainty and between heterogeneous partners identified trust as a crucial condition for positive outcomes, as it allows partners to exchange knowledge

with less concern about the counterpart's intentions and behaviors (Cullen, Johnson, and Sakano, 2000). Therefore, it can be expected that trust facilitates both the transfer of scientific knowledge to company researchers and innovation-related performance, such as patent applications and the quantity and quality of product innovations.

H4a: Trust is positively related to knowledge transfer in UI collaborations.

H4b: Trust is positively related to innovation performance in UI collaborations.

Methodology

Sample and Data Collection

The hypotheses are tested with survey data from U.S. firms active in UI research collaborations in the biotechnology industry. A company list obtained from the Lexis-Nexis Research database complemented by the directory of the Biotechnology Industry Association contained 660 firms with their biotech research dedicated to the life sciences.

Preliminary informants in each firm were contacted by phone and instructed in detail about the survey. The ultimate key informants were carefully identified and screened. These typically were the chief scientists or CEOs of the respective dedicated biotechnology company. To control for common method bias, the survey was split into two parts. The first part contained questions related to the independent and control variables, whereas the second part, which was sent out upon return of the first part, contained the dependent variables. Moreover, normative wording such as “trust” or “success” was avoided in the survey instrument to minimize the potential for social desirability bias among respondents.

All 660 firms were contacted by phone. One third of the firms reported no UI collaboration within the prior 3 years and was excluded from further consideration. Of the remaining 440 firms, 112 key informants provided responses from as many biotechnology companies in the life sciences, resulting in a response rate of 25.5%. Seven responses with missing data were excluded, resulting in a final sample of 105 complete responses.

The unit of analysis for this study is the most recent UI research collaboration project, completed within the last 3 years, which involved R&D of a new product or technology.

Characteristics of the sample firms are summarized in Table 1. The firms have a median number of 40 employ-

Table 1. Characteristics of Sample Firms

	Median	Standard Deviation
Number of employees	40	8,643.59
R&D workforce (full time equivalents)	12	726.18
Age of firm (years)	11	36.64

n = 105.

ees, a median R&D workforce of 12 full-time equivalents, and a medium age of 11 years. Eighty-one percent of the sample firms are SMEs with less than 500 employees. Overall, the sample mostly contains young research-intensive firms which are typical for the biotechnology industry. It broadly covers dedicated biotechnology firms from all regions of the United States. Moreover, 71.2% of all UI champions in the sample hold executive positions in their companies.

Measures

All key variables in this research were measured with 7-point multi-item Likert scales or semantic differentials (see the Appendix), which were adapted from previous studies whenever possible.

Dependent variables. *Trust* was measured with five items adapted from Ganesan (1994) related to the perceived frankness, reliability, honesty, and benevolence of the university partner. *Knowledge transfer* was a newly developed scale and measured with three items the absorption of new scientific knowledge, the discovery of new R&D themes, and the training of company researchers through the UI collaboration. *Innovation performance* was operationalized with three items adapted from Santoro and Bierly (2006) and gauged the amount of new patent applications, and the amount of newly developed products resulting from the UI collaboration, and the newness of newly created products to the market.

Independent variables. *IP policies* were assessed with a four-item scale adapted from Santoro and Bierly (2006), capturing the extent to which the university partner's IP policies were clear, flexible, and equitable, and the IP office was helpful in matters of technology transfer. *Shared governance* was measured with five items adapted from Heide and Miner (1992) that assess the extent to which the collaboration partners conducted joint planning, joint evaluation of the UI project, jointly agreed on project adjustments, and shared responsibility for the project. *Champion behavior* captured the extent to which

the UI champion overcame obstacles, did not give up when faced with opposition, got the right individuals involved, and got problems related to the collaboration into the hands of those who can solve them. This four-item scale was based on Howell, Shea, and Higgins (2005).

Control variables. *Company size* was included to account for any company specific effects and measured by the (log) number of employees for the focal firm. *Project length* was measured in months. Another variable captured the character of the collaboration project as a *basic research project* (coded 1) versus applied or development oriented. *UI collaboration experience* was gauged with a direct question on the amount of projects the company had previously conducted with the partner university, ranging from 1 (= none) to 7 (= a lot). Finally, the *UI collaboration importance* for the focal company was measured on a scale from 1 (= not very important) to 7 (= highly important).

Reliability and Validity

Convergent validity is examined by means of Cronbach's alphas and composite reliabilities (Appendix). All Cronbach's alphas (except for knowledge transfer) and all composite reliabilities are $> .70$, a benchmark for reliability (Hulland, 1999; Nunnally, 1978). The average variance extracted (AVE) and the square roots of each AVE are calculated to test for discriminant validity, as suggested by Fornell and Larcker (1981). The AVE exceeds .50 in all cases. The square roots of the AVE for each construct exceed the correlations between these and other variables (Table 2). Furthermore, a common factor

analysis of all items related to the main variables of the study (IP policies, shared governance, champion behavior, trust, knowledge transfer, innovation performance) settles on a six factor solution with all loadings $> .50$ and all cross loadings $< .40$. Taken together, the scales have satisfactory convergent and discriminant validity (Fornell and Larcker, 1981).

Results

Table 2 summarizes the descriptive statistics for the variables used in the study. The data are analyzed with partial least squares (PLS) estimation using Smart PLS 2.0 (Ringle, Wende, and Will, 2005). This estimation method allows the simultaneous estimation of measurement and path coefficients for relatively complex models and modest sample sizes and is a preferred method for exploratory research, such as this study on university–industry collaborations (Hair, Ringle, and Sarstedt, 2011; Reinartz, Haenlein, and Henseler, 2009). As PLS is a distribution-free estimation method, a bootstrapping resampling estimation with 500 samples was performed. Table 3 reports the results of the structural model estimations. The one-tailed significance of statistical associations is used, as PLS tends to conservatively underestimate the strength of paths in structural models (Reinartz et al., 2009).

IP policies ($\beta = .21$; $p < .05$) and shared governance ($\beta = .25$; $p < .01$) are both positively related to trust. Thus, H1 and H2 are supported. The interaction of champion behavior with IP policies is negatively ($\beta = -.23$; $p < .05$) and with shared governance is positively ($\beta = .23$; $p < .01$) associated with trust, lending support to H3a and H3b. Trust is positively related to knowledge

Table 2. Correlations and Descriptive Statistics

	1	2	3	4	5	6	7	8	9	10	11
1. Firm size	(n/a)										
2. Project length	-.02	(n/a)									
3. Basic research	-.01	-.07	(n/a)								
4. UI collaboration experience	.20*	-.06	-.00	(n/a)							
5. UI collaboration importance	-.23*	.26**	-.16	.10	(n/a)						
6. IP policies	-.06	-.07	.12	.17	.06	(.76)					
7. Shared governance	.17	-.08	.03	.10	.17	.08	(.76)				
8. Champion behavior	-.08	.06	-.07	.16	.11	.18	.21*	(.78)			
9. Trust	.13	-.16	-.08	.21*	.07	.33***	.33***	.23*	(.86)		
10. Knowledge transfer	.04	-.15	.04	.29**	.28**	.16	.22*	.36***	.37***	(.77)	
11. Innovation performance	-.26**	.22*	-.17	.17	.43***	.17	-.08	.26**	.16	.34***	(.78)
Mean	4.25	33.43	.24	3.69	4.87	4.95	5.02	5.45	5.24	4.36	3.40
Standard deviation	2.47	24.32	.43	1.99	1.64	1.30	1.25	1.09	1.17	1.38	1.68

* $p < .05$; ** $p < .01$; *** $p < .001$ (two tailed); $n = 105$; numbers on the diagonal are AVE square roots.

Table 3. PLS Path Coefficients

	Dependent Variable		
	Trust	Knowledge Transfer	Innovation Performance
<i>Main effects</i>			
IP policies (IPP)	.21* (2.16)		
Shared governance (SG)	.25** (2.56)		
Champion behavior	.07 (1.08)		
Trust		.28** (2.68)	.16* (1.79)
<i>Interaction effects</i>			
Champion behavior * IPP	-.23* (2.21)		
Champion behavior * SG	.23** (2.76)		
<i>Controls</i>			
Firm size	.16* (2.05)	.05 (0.71)	-.25** (2.78)
Project length	-.18* (2.06)	-.16 (1.49)	.18* (1.91)
Basic research	-.18* (2.23)	.10 (1.36)	-.10 (1.39)
UI collaboration experience	.06 (0.97)	.17* (2.02)	.18* (2.01)
UI collaboration importance	.05 (0.75)	.32*** (3.91)	.28** (3.04)
R ²	.39	.28	.30

* $p < .05$; ** $p < .01$; *** $p < .001$ (one tailed); β coefficients (t values); $n = 105$.

transfer ($\beta = .28$; $p < .01$) and innovation performance ($\beta = .16$; $p < .05$). Therefore, H4a and H4b are supported as well.

The control variables are also partially related to trust and UI collaboration outcomes. Firm size is positively related to trust ($\beta = .16$; $p < .05$) and negatively related to innovation performance ($\beta = -.25$; $p < .01$), whereas project length is negatively associated with trust ($\beta = -.18$; $p < .05$) and positively related with innovation performance ($\beta = .18$; $p < .05$). Basic research is negatively related to trust ($\beta = -.18$; $p < .05$). Finally, UI collaboration experience and UI collaboration importance are both positively associated with knowledge transfer ($\beta = .17$; $p < .05$ and $\beta = .32$; $p < .001$) and innovation performance ($\beta = .18$; $p < .05$ and $\beta = .28$; $p < .01$).

Discussion

This research set out to examine the relevance of IP policies, shared governance, and champion behavior for trust formation in UI collaborations. The results of the study indicate that all three factors play an important role for removing some of the roadblocks to successful R&D partnering.

First, trust formation in UI collaborations is supported by the flexibility and transparency of universities' IP policies. This finding illustrates that the degree to which such policies are being perceived as clear, fair, and flexible by the company partner in fact strongly determines the quality of the relationship between the collaborators, at

least in the U.S. biotechnology industry, where disagreements and conflicts over IP policies appear to be widespread (Kramer, 2008).

Second, shared governance of UI collaborations by firms and universities also contributes to the amount of trust created between the partners. This result indicates that joint planning and responsibility and joint evaluation and adjustment of the collaboration project help prevent misunderstandings and promote effective teamwork not only in R&D collaborations between firms (Asmus and Griffin, 1993; Lawson et al., 2009) but also in UI collaborations. Shared governance also allows responding to changes in the project environment in a coordinated and mutually agreed way.

Third, champions can separate ongoing discussions of initial IP agreements from the more important task of cultivating a mutually productive working relationship (Burnside and Witkin, 2008). Champion behavior reduces the relevance of IP policies and amplifies the importance of shared governance for trust formation. Because UI champions usually are located high up in the hierarchy, they can effectively motivate collaboration partners to adapt and redefine their working goals and plans should the environment change over the course of the project. This finding suggests that by promoting the collaboration, UI champions tend to redirect the focus of attention of the researchers and managers involved in the project from formal policies and rules to ongoing mutual efforts to get the job done and make the collaboration a success.

The results also indicate that trust formation in UI collaborations is negatively related to project length and basic research orientation and positively related to firm size. It is conceivable that the longer the project lasts the more the industry partner becomes impatient with the collaboration progress as university researchers are usually less affected by timeliness and financial bottom lines than their corporate counterparts. A company might interpret such a situation as a lack of commitment of the other side resulting in conflicts or misunderstandings that negatively affect trust. In a similar vein, due to the high failure rate of basic research projects in the biotechnology industry, frustration with less than hoped for outcomes may negatively influence the relationship between UI collaboration partners. The observed positive association between firm size and trust could express the higher confidence of larger firms in collaborating closely with universities.

Taken together, the results indicate that IP policies, shared governance, and champion behavior appear to be relevant mechanisms for developing trust in UI collaborations. At the same time, the relevance of these factors may vary between the stages of collaboration. The flexibility and transparency of IP policies primarily helps with finding initial agreements and successfully finalizing negotiations between companies and universities at the onset. Later on, the partners can focus on cultivating their relationship by joint planning, evaluation, and adjustment of their collaborative activities. UI champions help them shift their attention from formal agreements to effective working relations.

Trust is also strongly related to UI collaboration outcomes. In trusted collaborations, companies can absorb more scientific knowledge and tend to achieve better innovation performance. These findings illustrate the central role of trust for achieving success in UI collaborations. At the same time, however, the outcomes of UI collaborations are also related to other factors. When partners have prior UI collaboration experience and firms regard UI collaborations as important, companies learn more from projects with universities and achieve a stronger innovation performance. Thus, continuity and commitment in UI collaborations seem to be important. Larger firms are less satisfied with the innovation performance of UI collaborations than smaller ones. This observation could be related to the stronger internal research capabilities of larger firms, which set a higher bar for UI collaborations to deliver. Finally, project length is positively associated with innovation performance. Specifically, longer UI collaborations tend to result in more patents and more new products developed.

Limitations, Research Directions, and Managerial Implications

This study is based on the analysis of perceptions of one side of the UI collaboration only (the industry perspective), and on a single response for each collaboration. Attempts were made to collect the names of partnering universities. However, managers were extremely reluctant to provide such information. Thus, collecting dyadic data proved to be infeasible. Instead, by making an intense effort to identify the right informants and instruct them carefully on taking the survey, the quality of responses was maximized.

Survey data based on responses from single informants are potentially subject to common method bias. Various tactics regarded as effective remedies to control for such a bias (Podsakoff, MacKenzie, Lee, and Podsakoff, 2003) were taken, but its existence cannot be strictly ruled out.

Moreover, as this study is based on one-time observations, its potential in capturing the dynamics of UI collaborations over time is limited.

Given the contributions and limitations of this study, further research on the role of interorganizational governance mechanisms in UI collaborations in general and on the role of innovation champions in particular appear to be promising. Qualitative research based on case studies may complement this survey-based study.

Research oriented companies have significant concerns about IP and their ability to control it. Many universities would be hesitant to sign the agreement that companies use with employees or subcontractors to protect their IP but tend to insist on their own control. In addition, business models vary between industries, making it difficult for companies to accept a university's "one-size-fits-all" IP policy. As a consequence, many companies do not enter into R&D collaborations with domestic universities, even in the biotechnology field.

This study examined mechanisms that could alleviate some of these concerns and facilitate a trusting collaboration between industry and academia. In the United States today, adhering to IP policies is largely unavoidable for industry, but the policies are not necessarily one-sided and a detriment to collaboration. Rather, they form the framework within which the research collaboration unfolds. Universities have come a long way to offer flexible, clear, and equitable policies that cater to the needs and expectations of both sides, and companies should appreciate these efforts. Beyond these policies, an active and engaged participation in the research collaboration in the form of shared governance by the industry

partner is particularly critical for the cultivation of a trusting working relationship and the ultimate success of the collaboration. This shared governance also extends to higher level innovation champions that can mend fences if needed or amplify positive project governance. Thus, companies engaged in UI collaborations should focus on shared responsibility, planning, and adjustments with university partners and empower UI champions to help overcome collaboration obstacles.

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Appendix. Multiple Item Measures of Key Variables

	Factor Loading
Knowledge Transfer ($\alpha = .65$; composite reliability (CR) = .81; Average Variance Extracted (AVE) = .59)	
Through this UI collaboration . . .	
—new scientific knowledge was absorbed.	.66
—new future R&D themes were discovered.	.81
—in-house researchers were trained.	.83
Innovation performance ($\alpha = .71$; CR = .81; AVE = .63)	
As the direct result of this UI collaboration . . .	
—new patent applications were made.	.71
—new products were developed.	.83
—products that are completely new to the market were created.	.84
Trust ($\alpha = .86$; CR = .90; AVE = .65)	
—The university partner's representatives were frank in dealing with us.	.80
—Promises made by the university partner's representatives were reliable.	.86
—The university partner's representatives did not make unwarranted claims.	.77
—If problems (such as delays) arose, the partner's representatives were honest about the problems.	.89
—We felt the university partner's representatives were on our side.	.70
IP policies ($\alpha = .76$; CR = .85; AVE = .58)	
—University IP policies were clear and easily understood.	.72
—University IP policies were sufficiently flexible to meet our firm's needs.	.75
—The university intellectual transfer office was helpful with technology transfer.	.79
—University IP policies were equitable in revenue and royalty sharing.	.79
Shared governance ($\alpha = .82$; CR = .87; AVE = .58)	
—We were jointly responsible for getting things done.	.73
—We jointly planned how this project should be run.	.72
—Adjustments to project specific agreements were mutually agreed upon.	.73
—We shared responsibility to make sure that the relationship was mutually rewarding.	.88
—We jointly re-evaluated the progress of our partnership throughout the project.	.72
Champion behavior ($\alpha = .79$; CR = .86; AVE = .60)	
The person in our company who was most engaged in this UI collaboration . . .	
—showed tenacity in overcoming obstacles related to the UI collaboration within the company.	.76
—did not give up when others said this UI collaboration cannot be done.	.79
—got problems related to this UI collaboration into the hands of those that can solve them.	.75
—got the right people in our company involved.	.80